Classical Entailment for RuleTaker, CounterBench, and FOLIO

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Abstract

This report describes a reasoning system that decides whether a query follows from a knowledge base. The answer is True when the query holds in every situation that satisfies the facts and rules, and False otherwise. The three public datasets are not written in the same form, so each one is prepared in the way that matches it, and the last step is always a classical decision. RuleTaker is turned into ground Horn clauses and closed by forward chaining. CounterBench is turned into a twin-network theory and decided by fast resolution. FOLIO is read from the formulas supplied with the data and decided by the Z3 satisfiability solver. On the public data the operational accuracies are 98.46% for RuleTaker (1,625 stories), 94.58% for CounterBench (1,200 stories), and 76.82% for the binary FOLIO formulas (811 stories). Taken together, those three sets contain 3,636 stories and the accuracy is 92.35%. Counting FOLIO's Uncertain label as False, on all 4,029 stories, the accuracy is 92.53%. A solver result that is unknown stays Unresolved.

1 Introduction

The problem. The task is a yes-or-no question: given what is known, does this statement follow? The knowledge base is a set of facts and rules. The query is one statement. The answer is True when the query is entailed, which means it is true in every situation in which every fact and every rule is true. Otherwise the answer is False. The decision has to be produced by a procedure that can be checked, because a guess from a statistical model does not show that the premises force the answer.

Thought process. Once a story is written in logic, entailment is a precise question, and the procedure can be chosen for the shape of that logic. RuleTaker stories in this collection are variable-free if-then rules. The consequence relation is then the closure of a finite set of atoms, and forward chaining computes that closure completely. CounterBench stories ask what would happen after an intervention in a causal system. The factual world and the intervened world have to be kept apart, which a twin network does, and the resulting formulas are decided by resolution refutation. FOLIO stories are full first-order formulas, with quantifiers, exclusive or, and equality. Those formulas are sent to a satisfiability solver: the query follows exactly when the premises together with its negation have no model. A file that already contains formulas uses this same solver and skips translation.

Key contributions.

- **A dataset-specific pipeline.** English is translated only for RuleTaker and CounterBench, where the released text follows templates. FOLIO uses the annotated formulas as given.

- **Three classical procedures.** Forward chaining for ground Horn rules, fast resolution for the twin-network clauses, and Z3 for general first-order formulas.
- **An evaluation on the three public datasets.** Accuracy and reasoning time for every RuleTaker split, for CounterBench, and for FOLIO with Uncertain removed and with Uncertain counted as False, followed by one combined score.

2 Datasets

The three collections are the public sets named in the assignment. They are read from the local copies, in the released order. The sources are listed below.

RuleTaker. Source: https://huggingface.co/datasets/jise/ruletaker. Each example is a short English theory and a statement to check. The local files keep the released splits: train (1,000), test (250), and ood test (375), which is 1,625 stories in all. The sentences follow a small set of patterns, such as "Bob is young" and "If someone is young, then someone is smart". The label is the released flag, True or False. Of the 1,625 labels, 881 are True and 744 are False.

CounterBench. Source: https://huggingface.co/datasets/CounterBench/CounterBench. The local table has 1,200 counterfactual questions about made-up causal stories. A story names fictional events and says which events cause which others. The question asks whether an outcome would still occur after one or more events are switched off. An example has the shape "Ziklo causes Blaf. Would Blaf occur if not Ziklo instead of Ziklo?". The label is the released true or false answer.

FOLIO. Source: https://huggingface.co/datasets/yale-nlp/FOLIO. Each story is written by people and is paired with expert first-order logic for the premises and the conclusion, for example $\forall x\,(\mathrm{Dog}(x) \rightarrow \mathrm{Animal}(x))$. The source files contain train (1,001) and validation (203), which is 1,204 stories. The labels are True, False, and Uncertain: 460 True, 351 False, and 393 Uncertain.

The logic given in the dataset is used directly. A label counts as True only when the released label is True. In one experiment the 393 Uncertain stories are removed, leaving the 811 stories whose label is True or False (677 from train and 134 from validation). In the other experiment those 393 stories are kept and counted as False.

3 Implementation details

The notebook runs the three preparations below. Figure 1 shows the order: identify the dataset, build the logical form that matches it, then call the procedure that decides that form. Reading the files, and the translation step for the two English datasets, are not included in the reasoning time.

3.1 RuleTaker: templates, then forward chaining

A fact such as "Bob is young" becomes the atom Young(bob). A rule such as "If Bob is young, then Bob is smart" becomes

$$\mathrm{Young}(\mathrm{bob}) \rightarrow \mathrm{Smart}(\mathrm{bob}).$$

The rules in this collection have one atom in the body and one atom in the head. There are no function symbols and no variables, so only finitely many atoms can ever be derived.

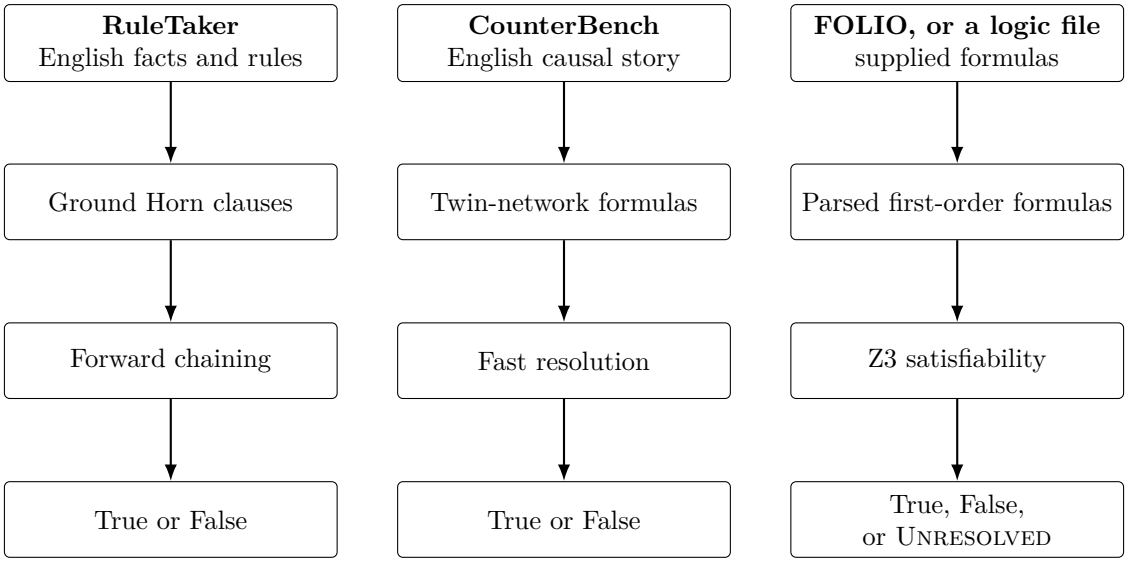


Figure 1: The sequence used for each dataset. A test file that already contains formulas follows the right-hand column.

Forward chaining starts from the facts. Whenever the body of a rule is already present, the head is added. The loop stops when a pass adds nothing. The query is entailed exactly when it appears in the resulting set. If the search finishes and the query is absent, the answer is False. That absence is a real answer for this fragment, because the closure is complete.

3.2 CounterBench: a twin network, then fast resolution

A CounterBench story is a causal system and a counterfactual question. "Ziklo causes Blaf" is a structural equation, and the question changes one event in an imagined world. The translator builds two copies of each event. OccurF(x) means that x occurs in the factual world. OccurC(x) means that x occurs after the intervention. Each causal statement becomes a biconditional. An intervention deletes the equation of the intervened event and sets that event false in the counterfactual world. Events with no cause keep the same value in both worlds. Observations constrain only the factual world. The query is the atom for the asked event in the counterfactual world.

Those formulas are converted to clauses. Implications and biconditionals are rewritten with conjunction, disjunction, and negation. Negation is pushed inward, existential quantifiers are Skolemised, and the result is a set of disjunctions. Entailment is decided by resolution refutation: the clauses of the negated query are added, and the search looks for the empty clause. A contradiction means the query is entailed. A finished search with no contradiction means it is not entailed.

The search is the fast variant. It uses the same resolution rule as plain resolution and adds three restrictions that drop pairs which cannot help:

1. A predicate index stores each clause under the predicate and sign of its literals, so a literal is paired only with the complementary predicate.
2. Unit subsumption drops a new clause when a known unit fact already makes that clause true.
3. Unit simplification deletes a literal whose complement is a known unit fact, which shortens the clause immediately.

The time limit is 0.3 seconds per query. If the search stops early, the prediction stays UNRESOLVED. On this dataset the fast search decides every story.

3.3 FOLIO and other logic files: Z3

The FOLIO files already contain the premises and the conclusion in first-order logic. Those strings are parsed and sent to Z3. Free variables in the premises are read universally, and free variables in the query are read existentially. When a formula asserts no positive equations, distinct constants are treated as distinct, up to a cap of 60 names. Predicate alignment is turned on for FOLIO, because some annotated predicates are spelled in more than one way. A test file of formulas uses the same call with alignment turned off, so names are kept as written. The parser also accepts the ASCII spellings `forall`, `exists`, `~`, `&`, `|`, `->`, and `<->`.

The solver tests the premises together with the negated query. An unsatisfiable problem means the query is entailed, and the prediction is True. A satisfiable problem means the query is not entailed, and the prediction is False. If the solver returns unknown, or the limit of one second is reached, the prediction stays UNRESOLVED. Two FOLIO experiments share this call. In the first, Uncertain examples are absent. In the second, Entailment is mapped to True, and both Contradiction and Uncertain are mapped to False. The source files are not edited.

3.4 An example trace

The premises

$$\text{Young(bob)}, \quad \text{Kind(bob)}, \quad \text{Young(bob)} \to \text{Smart(bob)}, \quad \text{Smart(bob)} \to \text{Successful(bob)}$$

are ground Horn clauses. Forward chaining adds Smart(bob) from the first rule. The second rule then adds Successful(bob). The query is in the closure, so the answer is True. The same formulas, sent to Z3 as a logic-file query, are also entailed.

4 Results

Each table reports operational accuracy: the number of predictions that match the label, divided by every story in that split. An UNRESOLVED prediction counts as incorrect. Where the solver decides every story, operational accuracy and accuracy on the decided stories are the same number. Reasoning time is the mean and the sample standard deviation of the solver call only. Reading the files, and translating RuleTaker and CounterBench from English, are left out of that time. Sections 4.1–4.4 keep FOLIO's Uncertain stories out. Section 4.5 puts them back, counted as False, and repeats the combined score under that labeling.

4.1 RuleTaker

Forward chaining decides all 1,625 stories. The train, test, and out-of-distribution splits stay within about two points of each other, which fits this collection: the out-of-distribution stories use the same fact and rule patterns, so the same closure still applies. The 25 misses are stories whose rules derive the query while the released label is False. The procedure does not consult the label when it adds a fact.

Split	Examples	Accuracy (%)
train	1000	98.30
test	250	97.60
ood test	375	99.47
All	1625	98.46

Table 1: RuleTaker. Forward chaining decides every story: 1,600 of 1,625 are correct. Reasoning time on the full set is 0.000013 ± 0.000003 seconds per story.

4.2 CounterBench

Split	Examples	Accuracy (%)
all	1200	94.58

Table 2: CounterBench. Fast resolution on the twin network decides every story: 1,135 of 1,200 are correct. Reasoning time is 0.0022 ± 0.0048 seconds per story.

Fast resolution returns a definite answer on all 1,200 stories: 570 queries are entailed and 630 are not. That agrees with the released label on 1,135 stories. The other 65 are stories whose twin-network theory and the released label disagree. The time limit of 0.3 seconds is never what stops the search here, so none of these rows are stored as UNRESOLVED.

4.3 FOLIO

Split	Examples	Accuracy (%)
train	677	75.92
validation	134	81.34
All	811	76.82

Table 3: FOLIO with Uncertain removed. Z3 is correct on 623 of 811 stories. Two queries stay unknown, so 809 stories are decided and the accuracy on those stories is 77.01%. Reasoning time is 0.0236 ± 0.0597 seconds per story.

This is the harder FOLIO setting, because every remaining label says that the conclusion definitely follows or definitely does not. The solver returns 325 entailed, 484 not entailed, and 2 unknown. On many gold-True rows the premises and the negated conclusion are satisfiable, so the answer False records that the conclusion does not follow from the formulas as written. The two unknown rows stay UNRESOLVED and count as incorrect in the operational accuracy.

4.4 Overall

The combined row counts each story once and uses the binary FOLIO set, so Uncertain stories are not in the 3,636. The accuracy is 92.35%, which is 3,358 correct predictions. RuleTaker and CounterBench together contribute 90 of the 278 misses. The other 188 come from the binary FOLIO set, including the two queries Z3 leaves unknown. The three procedures are different, so their times are left in Tables 1–3 and are not added into one total.

Dataset	Method	Examples	Correct	Accuracy (%)
RuleTaker	Forward chaining	1625	1600	98.46
CounterBench	Fast resolution	1200	1135	94.58
FOLIO	Z3	811	623	76.82
All three		3636	3358	92.35

Table 4: The three datasets together, with FOLIO's Uncertain stories removed. Each story is counted once.

Data	Uncertain	Examples	Accuracy (%)
FOLIO train	removed	677	75.92
FOLIO train	counted as False	1001	81.72
FOLIO validation	removed	134	81.34
FOLIO validation	counted as False	203	86.21
FOLIO all	removed	811	76.82
FOLIO all	counted as False	1204	82.48
All three	removed	3636	92.35
All three	counted as False	4029	92.53

Table 5: Removing FOLIO's Uncertain examples, or counting them as False. The full FOLIO run is correct on 993 of 1,204 stories. Three queries stay unknown. Reasoning time is 0.0297 ± 0.0620 seconds per story. The 811 binary stories inside this run again score 76.82%.

4.5 FOLIO: removing Uncertain or counting it as False

The full source has 460 True labels, 351 False labels, and 393 Uncertain labels. After Uncertain is counted as False there are 460 True labels and 744 False labels. The solver returns 347 entailed, 854 not entailed, and 3 unknown. Of the 393 Uncertain stories, 370 are answered False, 22 are answered True, and 1 stays UNRESOLVED, so 94.15% of them match the mapped label. An Uncertain conclusion is one that the premises do not prove. A solver that answers True only when the premises and the negated conclusion are unsatisfiable answers False on most of those rows, and that matches the mapped label. The three unknown predictions are still stored as UNRESOLVED.

The setting without Uncertain is the harder of the two, because it keeps only the conclusions marked definitely true or definitely false. Adding the Uncertain stories raises FOLIO from 76.82% to 82.48%, and raises the three-dataset score from 92.35% on 3,636 stories to 92.53% on 4,029 stories. The prediction file stores both FOLIO settings, so it has 4,840 rows. The combined scores above count each story once.

5 Conclusion

The system answers entailment with a classical procedure chosen for the form of each dataset. Forward chaining is complete for the ground Horn fragment and scores 98.46% on RuleTaker. Fast resolution on the twin network scores 94.58% on all 1,200 CounterBench stories. Z3 on the annotated FOLIO formulas scores 76.82% when Uncertain examples are removed. Across those three sets, 3,358 of 3,636 stories are correct, which is 92.35%. Counting Uncertain as False raises the FOLIO score to 82.48% and the three-dataset score to 92.53% on 4,029 stories. Unknown solver results remain UNRESOLVED.